UNIVERSITY OF HERTFORDSHIRE
School of Computer Science

BSc Honours in Computer Science

6COM1053- Computer Science Project

Final Report
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Food Maths
A diet and fitness application - to aid users in calorie counting.

M.C.Maillet

Supervised by Helen Partou

Abstract

The purpose of this project is to develop an application which can aid in solving the real-world problem of obesity and unhealthy lifestyles. The application achieves this by focussing on minimalistic although meaningful functionality as well as simplicity in design to facilitate users in achieving their own desired health goals. The key focus is on efficiency and ease of use, as these are the defining factors in user retention for such applications.

Acknowledgements

Before I begin my report, I would like to direct my gratitude and thanks to Helen Partou my project supervisor, for helping me stay on track with continues guidance and motivational emails.

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1. Introduction

1.1 Project Overview

As mobile phones have become ever more prevalent in todays society, we have seen a shift in the way people are keeping track of their daily lives. Whereas in the past professional athletes would have gone and seen a diet and nutrition specialist costing upwards of £50 per hour, many are now turning to modern technology to provide them with the same information and assistance they require. In the past decade alone, there has been an influx of fitness and diet tracking apps, these apps appeal to a wide array of users not only the proffesional athletes, as they are a very affordable if not a completely free alternative to seeing a specialist.

App name (year established)	Producer (country of origin)	Rating (Android/iOS/ratings)	Downloads (Android)[a]
Calories Counter (2018)	FatSecret (Australia)	4.4/4.5/1300	10,000,000
LifeSum (2018)	LifeSum (Sweden)	4.3/4.5/22,600	5,000,000
MyPlate Calorie Tracker (2017)	Livestrong (United States)	4.6/4.5/7900	500,000
Argus (2015)	Azumio (United States)	4.3/4.5/22,600	100,000
Lose It! (2008)	FitNow (United States)	4.4/4.0/156,600	5,000,000
Calorie Counter & Diet Tracker (2009)	MyFitnessPal (United States)	4.6/4.5/490,000	50,000,000
MyDietCoach (2014)	InspiredApps (United States)	4.4/4.5/5700	10,000,000

Fig. 1. Popular diet and fitness app based on Android figures (Ferrara et al., 2019)

The table above shows the popularity of these applications, with the majority reaching over 5 million downloads the most being MyFitnessPal at 50 million. These numbers are not inconsequential and show a large interest in such applications.

1.2 Project Aim

My aim going into this project is to develop a fitness and diet tracking application, that could compete with current applications on the market and improve on them in some aspect. I plan to do this by building on the foundations of what they provide and adding my solutions to problems I have observed while undertaking my own research into the current available applications and their functionality.

1.3 Project Goals

My motivation for this project is to apply and develop the concepts I have learnt throughout the software engineering modules and produce an artefact that not only performs well, it also aids in addressing a global need/want.

My personal use of the MyFitnessPal application during the past three years has sparked my interest in developing a similar app concept with my own ideas and knowledge.

From my previous years module 5COM1052 Database Concepts, I plan to apply knowledge gained with additional learning I will undertake to design and implement a working backend database with the ability to handle SQL queries and integrate this into my application.

Since I am designing this app single-handedly and I am under time restrictions I am aware the application will not be up to the standard of the currently available applications on the market, however the intent of my project is to arrive at a working artefact which with further development could be a competitor for these other applications.

2. Background Research

In my research I have evaluated literature sources relevant to the project I plan to undertake. To best aid in the success of my application, I will need to be successful in key areas including: user retention, appropriate UI and design, and accurate accessible information. To complete my review, I will be analysing two popular applications currently available on Android as well as Jakob Nielsen's 10 general principles for interface design.

2.1 Evaluating Existing Applications

2.1.1 MyFitnessPal

I will start my review with an analysis of the MyFitnessPal app, with over 50 million downloads on Android at the time of writing this review, it is by far the most popular fitness and nutrition app on the market. The application is free to download, although they also offer a premium paid subscription service with additional features. The main purpose of the application is for calorie counting which is done by logging food consumed over the course of a day, if this is completed consistently throughout a week it can give users a deeper look into their own nutrition and diet habits. The application allows users to be more accountable and aware of how their daily food intake can affect them in the long-term and if any changes should be made to their current diet.

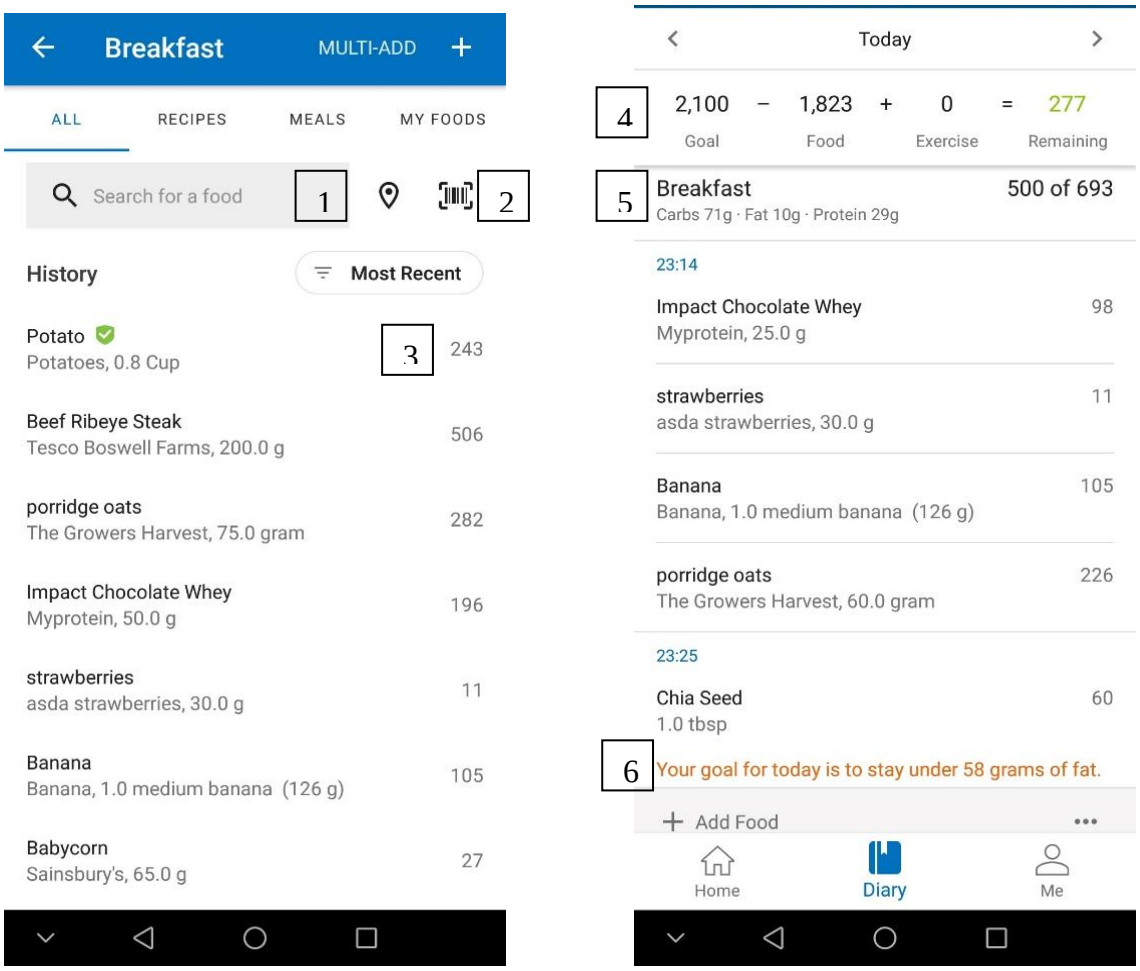


Fig. 2.

The two screen captures shown above are from the Android version of MyFitnessPal (MyFitnessPal Inc, 2019) with the premium subscription. The first screen capture shows the food search screen while the second shows a completed meal log.

A breakdown of the key points on each screen:

1. Search the user created food database with key words to find food items.
2. Barcode scanner - used to scan food packaging to speed up searches.
3. Displaying a previously consumed food item, along with portion size and calorie content.
4. Daily calorie goal, along with calories consumed and the net calories remaining.
5. A breakdown of the macronutrients consumed for breakfast and displayed further to the right are the calories consumed out of the total calories allocated for that meal.
6. A tip under a food entry reminding the user of their goal.

When it comes to the app's functionality, it is very simple to use, once installed users can quickly get started with their own account complete with weight and nutrition goals. The ability to bring in users and provide them with a personalised account as quickly as possible, is a feature MyFitnessPal uses to its advantage, as once a user has created their own account, they are much more likely to revisit the app. This is something I will aim to implement into my own applications design.

As shown in the screen capture there is a barcode scanner, the addition of this feature allows users to input food purchased almost instantly. By reducing the time taken to log food, users will be more likely to fill in their food log on a regular basis as it is of greater convenience. This would then lead to a more complete food diary and an improved display of information for the user, which could aid our users in achieving their health goals. However due to the limitations of my project it is unlikely I will be able to implement a feature of this nature, instead I will attempt to replicate the benefit it provides by keeping a focus on the convenience and speed of food logging.

2.1.2 Lifesum

The second application I will be analysing is Lifesum, another widely used fitness and nutrition application with over 10 million downloads on Android at the time of writing this review. Lifesum is a subscription only service and unlike my MyFitnessPal has no free functionality. Although the purpose of this application is also to aid in tracking calories the big difference between this app and MyFitnessPal, is that Lifesum offers a completely user tailored meal plan based on a questionnaire that the user may complete. This questionnaire considers food preferences along with allergies and any dietary requirements users may have.

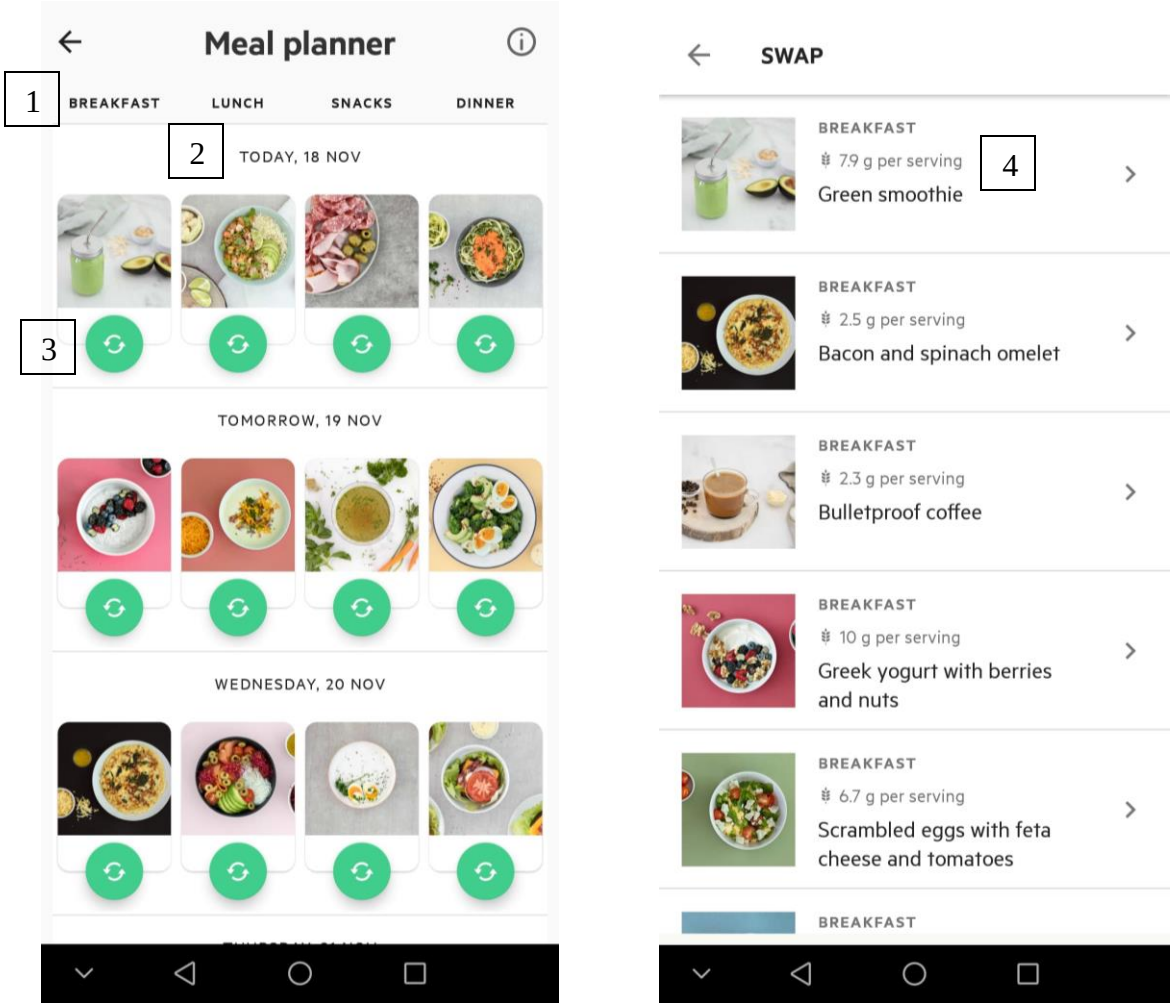


Fig. 3.

The screen captures above are from the Android version of the Lifesum application (Lifesum, 2019), the first capture is of a user tailored meal plan for 3 days and the second shows the user swapping a breakfast item. These captures were taken with a ketogenic meal plan preference.

A breakdown of the key points on each screen:

1. Showing the meal columns and how the food is split across 4 meals per day.
2. The date of the specific meal plan.
3. A swap button if users are not pleased with a meal option, they can swap it out for another suggestion still within the same dietary preferences.
4. The swap selection screen including multiple options to replace the current breakfast selection.

The tailored meal plan is by far Lifesum's best feature, allowing users to choose which foods they wish to have included in their meal plan (e.g. vegan, vegetarian or pescatarian) as well as the removal of any potential allergens (e.g. nuts, fish or eggs). This provides users with a meal plan consisting specifically of only meals they can and want to eat. This feature greatly benefits users who may not be experienced with meal planning and nutrition, as it takes away most of the difficulty that comes with transitioning to eating and sticking to a new diet. As the user is presented with a great variety of meals and can plan up to a month ahead at one time, this can help users to add structure into their daily routine. I plan to develop a meal

suggestion algorithm like that used by Lifesum for my application, as I believe this would assist users who are new to nutrition and meal planning, and therefore encourage them to uphold their goals to completion.

Like MyFitnessPal, Lifesum is very simple to setup and begin using. When it comes to the overall display, Lifesum has a cleaner and more professional finish than MyFitnessPal, however It comes at a cost. Much of the dietary information and user statistics are kept in specific places, this can at times mean that users must look through many different menus to find the specific information they are searching for. When first getting their bearings with the app, users may find this a little overwhelming especially if they are more interested in the nutritional statistic breakdown of a food then the meal planning service. For this reason, when designing my application, I will attempt to strike a balance between both applications, focussing on nutritional information being clear and easily accessible, as well as ensuring it is displayed in a clear and professional format.

A final noteworthy point comparing these two applications visually, specifically the colours they are using on many of their menus. MyFitnessPal uses only two colours for its entire interface which are blue and white. Lifesum similarly uses only two main colours for most of its interface, green and white. The symbolism of the colours used may be specific to health and fitness as the colour green is known to symbolise nature and freshness, while blue symbolises tranquillity and balance. While deciding on the colour scheme for my application I will have to consider the message that I would like to put across with the use of specific colours and visual cues.

2.2 Nielsen's Heuristics

User experience is a large focus when developing an application, for this reason I will be analysing the ten usability heuristics by Jakob Nielsen (Nielsen, 2019), these are his evaluations of the user experience for a digital platform. These evaluations are over 20 years old now, and they are still used to this day as a general baseline.

1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use Accelerators
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

Although all of these are applicable to the design of my application, I would like to focus on a select few. Number 3, User control and freedom – states that users often make mistakes when using systems and for this reason systems should be designed with a ‘undo’ functionality. This is intended to promote users to not feel timid when using a system and be able to explore all the possible areas and functions the system has to offer without the fear of making a mistake which cannot be undone. Whilst using both MyFitnessPal and Lifesum, I have not come across any method of undoing mistakes without going back onto the specific menus and fixing them manually. In my application design, I will aim to include an instant

way for users to undo mistakes made, without any additional navigating back through a menu.

Recognition rather than recall, heuristic number 6 – states that UI should facilitate memory recognition rather than memory recall, since recognition is easier for users and requires less cognitive function. By using a clear graphical interface with images and recognizable symbols we can aid the users' journey through an application, as they will have an easier time navigating the various menus and screens. This can be applied to my app, when it comes to designing the UI of my application, I will intend for users to have some initial expectation of the functionality of certain button presses, and what information will be displayed and available to them if they navigate to a specific screen.

Heuristic 7, flexibility and efficiency of use accelerators – states that successful UI design includes accelerators which can be used by experts to speed up the very basic tasks while not overwhelming the newer inexperienced users. This can be seen in both of the applications I have analysed earlier, e.g. during the food logging process on MyFitnessPal, it is possible for more experienced users to create 'meals' and save them for future use. If the user wishes to log the same meal again, they can simply add the full meal with all its ingredients at once, saving the experienced user valuable time, while not interfering with the new user's ability to learn the basics of the application. As I am aiming to reduce the length of time it takes users to log their food, my application will require a feature along the same lines as the functionality from MyFitnessPal.

From the sources I have evaluated I now have a much deeper understanding and plan in mind for the functionality and design aspects that would be best to implement into my application. To ensure the overall UX is enjoyable, simple and informative, I need to consider each of the points raised in this review. I will focus on creating an efficient user registration and simple to use food logging system to minimise time spent by the user upon each visit to my application, in an effort to increase the convenience for users to adopt my application as a part of their daily habits. Also, I intend on developing a user tailored meal suggestion algorithm which will now consider aspects I had not otherwise thought about such as food allergies, therefore expanding the total possible user base of my application. For users to self-evaluate effectively my application will require easily accessible and informative user statistics, which are not hidden in layers of menus. When considering each new iteration of my design and UI, I will need to prioritise UX in all aspects of my application, analysing the impact of each new feature as and when it is added with the use of Nielsen's heuristics.

3. Analysis

3.1 User Profile

To better aid me in developing my system requirements I have chosen to create user profiles, with the aim to provide a greater user experience within the application as I will make design decisions based on the users perceived needs and wants. “Having a fictional person (persona) representing a profile grounds the design effort in real users.” (User Modeling: Developing Personas and User Stories, 2020).

In order to reduce bias, I have decided against designing a persona based on myself and have instead opted for proto personas. This development was undertaken so that I may gather the information I require to decide on the functional and non-functional requirements that will be included in the design of my project.

These personas have been designed with a wide user base in mind ranging in ages, gender, and varying levels of fitness and computer literacy.

James Dickson	
Description • Male • 18 years old • Store assistant • Wants to keep track and stay in shape	Behaviour • Low disposable income • Independent • Very computer literate • Goes to the gym multiple times per week
Pain Points • Easily distracted • Easily frustrated by errors	Needs • Responsive functionality • Reliability throughout process • Minimalistic design

Fig.4.

Laura Jackson	
Description • Female • 48 years old • Team Manager • Looking to start tracking and transition to a healthier lifestyle	Behaviour • High disposable income • Highly connected • Busy lifestyle • Not technological adept
Pain Points • Can miss key information • Not a lot of free time to spend using the application	Needs • Simplicity in the design • Fast and efficient process • Security

Fig.5.

David Miller

Description	Behaviour
• Male • 26 years old • Professional athlete • Needs to be in the best shape of his life	• High disposable income • Obsessive • Determined • Has tried using apps in the past but he has not stuck to it
Pain Points	**Needs**
• Requires professional level accuracy • Gets lost in the menus of complex apps • Has only little time to spend	• Accurate information • Fast and efficient process • Minimalistic design

Fig. 6.

Now these personas have been created I can apply them to my project plan to decide on the specific requirements I will be focussing on when developing my app.

3.2 Use Cases

Use case diagrams are used to show interactions between actors (the system users) and the system itself with all its processes. They are used mainly to ensure the proposed design can deliver the required functionality and whether the system is dependable. The scenarios I have created are intended to describe a story of a possible user journey through my application, with the aim of providing me with additional information to assist me with requirement discovery.

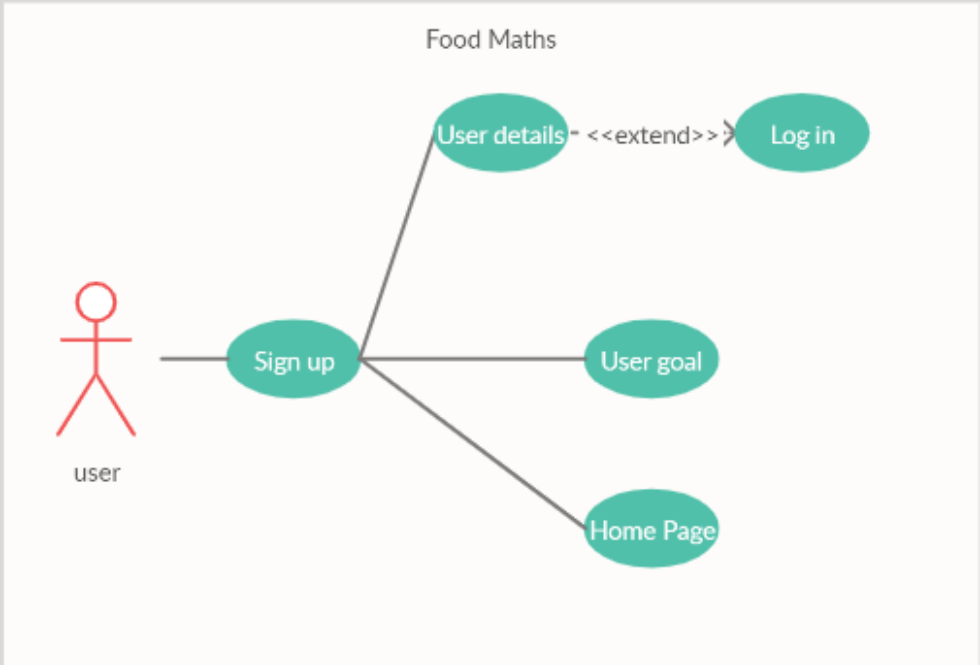


Fig. 7.

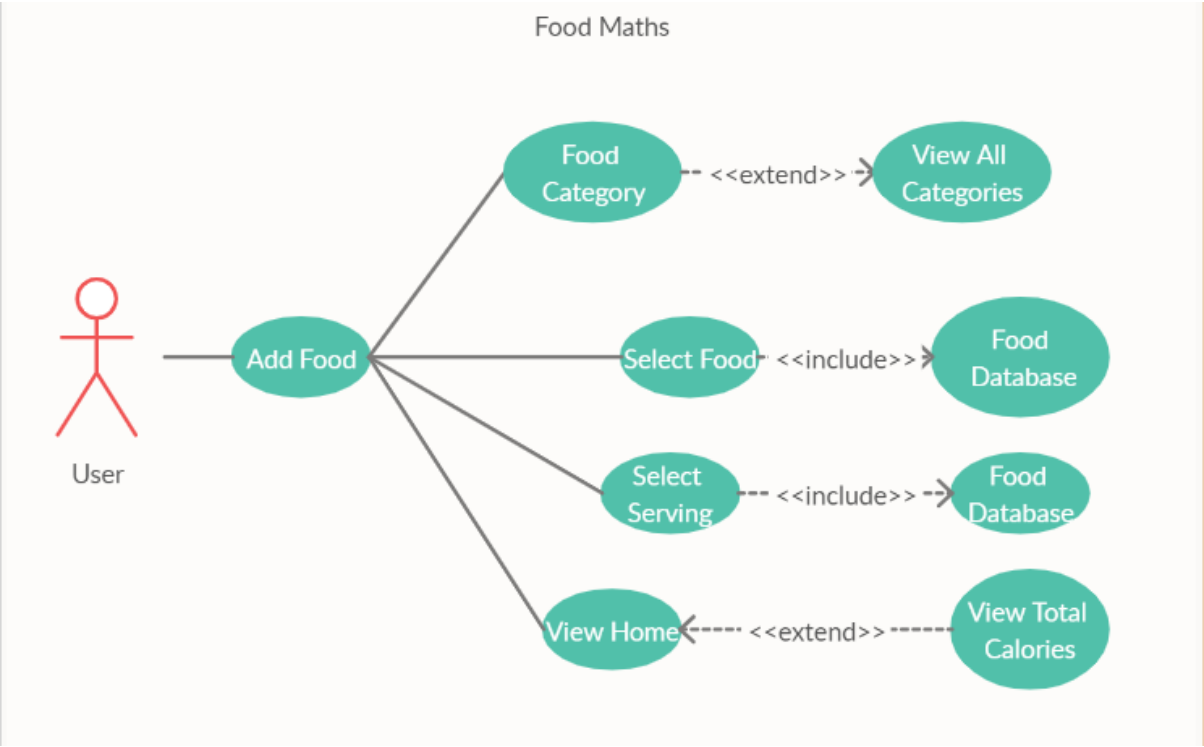


Fig. 8.

The use cases shown in Figure 7 and Figure 8 describe two stories, the first describes a user starting the application for the first time and therefore being prompted to sign up, the system checks if the user is already on the database and then proceeds to take user details. The user then defines their goals and is taken to the home page of the application where they can begin their activity.

The second use case shown describes a user who is already logged onto the system, trying to add a food item to their diary. The system allows the user to view all the food categories they have already saved and choose a food from the database. Once the food is selected the user defines the amount they are wanting to log into the diary. The system then calculates the total calories this will account for, and redirects the user back to their home page, where they can view the recently added diary entry and their total remaining calories for the day.

3.3 Functional Requirements

Functional requirements are a description of the intended service(s) that the software must offer. They are used to describe entire systems or their components, ranging from data inputs to business processes and user interactions. They are used to lay out in simple terms how the system should perform. Gathering these requirements is a key part of the project development cycle as without them it can be difficult to asses the success or failure of a project.

No.	Functional Requirement
1.	Users must be able to sign up
2.	Users must be able to choose a goal
3.	Users should be able to edit personal details
4.	Users should be able to edit their goal
5.	Users should be able to create new food
6.	Users should be edit food

7.	Users should be able to delete food
8.	Users should be able to add food to food diary
9.	Users should be able to edit food in food diary
10.	Users should be able to delete food in food diary
11.	Users should be able to see a display of total calories eaten
12.	Users should be able to see a display of calories remaining
13.	Users should be able to list foods by categories
14.	Users should be able to create new categories
15.	Users should not be able to leave inputs blank on sign up or sign up goal screen
16.	Users should be able to navigate the artefact using clearly labelled buttons

Fig. 9.

The functional requirements listed above describe the main functionality I intend to add into my application for it to be fully functional and the users to have the best experience possible.

The use of CRUD within the database is to ensure the model is complete and able to handle all the requests is may receive from users. "The CRUD paradigm is common in constructing web applications, because it provides a memorable framework for reminding developers of how to construct full, usable models." (What is CRUD? | Codeacademy, 2020)

3.4 Non-Functional Requirements

Non-functional requirements are used by developers to specify how a system should behave, and these requirements are used as constraints for the system to be judged against once development has finished.

No.	Non-Functional Requirement
1.	If the artefact encounters a bug it must provide the user with a message
2.	Artefact should be able to operate across a wide range of devices
3.	Artefact should be responsive
4.	Artefact should inform users of its current state
5.	Artefact should be able to hold multiple pieces of data in its database

Fig. 10.

These requirements are focused on ensuring the system behaves in an appropriate way once development has finished. For the purpose of testing the final artefact I will have a reference to look back on and evaluate how successful my artefact has been at achieving these requirements.

3.5 Stakeholders

Users of my application – They will be directly affected by any changes made to the application which may benefit them or put them at a disadvantage.

Customer support – Once we have an established user base, customer support will be required to maintain customer accounts.

Developers – Further development would be conducted by a developer or a team of developers as the application scope increases over time.

Potential investors – They may wish to invest if they see potential in my application, once invested they will benefit if the application is profitable.

App store/Google play store – In future the application could be available for download via either or both services, they would benefit from the increase in user visits.
Competitors – Depending on the success of my application they may experience a decrease in their customer base, alternatively they may update their applications to implement some of the features available on mine.

Partner systems – In future my application may link user accounts to partner systems for example social media such as Facebook. This would benefit the partner system as well as my own as our users would benefit from the cross over and additional features.

3.6 Risks

During the development of my artefact there is the opportunity for issues to arise and things to go wrong. To minimise and hopefully prevent this I will be going over some of the possible implementation risks, these may occur during the implementation or due to the implementation of my artefact.

3.6.1 Risk during implementation

Risk	Likelihood	Consequences	Avoidance
Gold plating	Moderate	Adding unneeded functionality and wasting time	Focus on functionality, requirements scrubbing, prototyping and cost-benefit analysis
Developing the wrong functions and properties	Low	Wasted time and resources spent	Organizational analysis, mission analysis, operations-concept formulation, user surveys and user participation, prototyping
Developing the wrong user interface	Low	Wasted time and resources spent	Prototyping, scenarios, task analysis, user participation
Personal shortfalls	Moderate	Inadequate skills or time	Staffing with top talent, job matching, team building, key personnel agreements, cross training
Unrealistic schedules and budgets	Moderate	Falling behind schedule	Detailed milestone cost and schedule estimation, design to cost, incremental development, software reuse, requirements scrubbing
Maintaining stakeholder involvement	Low	Developing an application which does not fulfill the needs required	Regularly review the developed application and compare it to the pre-defined solution

Continuing stream of requirements changes	High	This would add extra development time on to the current plan	High change threshold information hiding, incremental development

Fig. 11

3.6.2 Risk due to implementation

Risk	Likelihood	Consequences	Avoidance
Data inconsistencies	Moderate	Data with inconsistencies can cause a website deployment to fail - This would affect users and stakeholders as the system would be down and share prices could be affected	Data integrity
Availability disruptions	Moderate	Adding new code may cause the current system to go down or crash - This would affect users, stakeholders and customer support as the system would be down and share prices could be affected	Prototyping and thorough testing for each new iteration
Information security vulnerabilities	Low	Implementing a new feature that contains vulnerable information security that leads to widespread security attacks - This would affect users and stakeholders as the system would be compromised and share prices could be affected	Defending information from unauthorized access, prototyping, cryptography
Deployment process failure	High	Processes to deploy a system fail and the changes made need to be reversed – This would affect developers, as they must take the steps necessary to fix the issue	System development timing, configuration and process alignment, communication and support
Partner integration issues	Moderate	New services are launched and fail due to integration issues with a partner, the system must be pulled as the partner lacks the ability to resolve the issue on their side – This would affect our partner systems and developers, the users on our partners system would be unable to access the service, and our developers must work to fix the issue.	Continuous integration, prototyping, process integration and virtualization

Fig. 11

4. Development Tools

4.1 Android Studio

I have chosen to use Android Studio as my IDE, Android Studio has improved on the Eclipse ADT and is currently the official development platform for Android Applications, therefore it is the most up to date and supported platform for android application development. It has a large community of developers who can share ideas and concepts with each other, this has helped me at times when I was unsure why my artefact was not running as intended.

Android Studio facilitated my artefact development by providing me with an organised IDE, and a clear user interface. Coming from a position of never having used this IDE beforehand it did take me a few attempts to get started, however once I had my artefact running in the most basic form, I was able to run some initial tests which helped me grasp the additional features of Android Studio and use them to their full potential.

While developing with Android Studio I had to decide which API I wanted to develop for, after doing some research I came to the decision that android API 26 known as Oreo 8.0 would be the most suitable API to design my application for. The reason for this decision is that at this current point in time Oreo can be run on 60.8% of android devices, this statistic is visible to all developers on Android Studio. This meant my application would be usable by a large proportion of the Android user base and as Oreo was released in 2017, I would still be working on a up to date API with almost all the current features available.

Android Platform/API Version Distribution

ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99.8%
4.2 Jelly Bean	17	99.2%
4.3 Jelly Bean	18	98.4%
4.4 KitKat	19	98.1%
5.0 Lollipop	21	94.1%
5.1 Lollipop	22	92.3%
6.0 Marshmallow	23	84.9%
7.0 Nougat	24	73.7%
7.1 Nougat	25	66.2%
8.0 Oreo	26	60.8%
8.1 Oreo	27	53.5%
9.0 Pie	28	39.5%
10. Android 10	29	8.2%

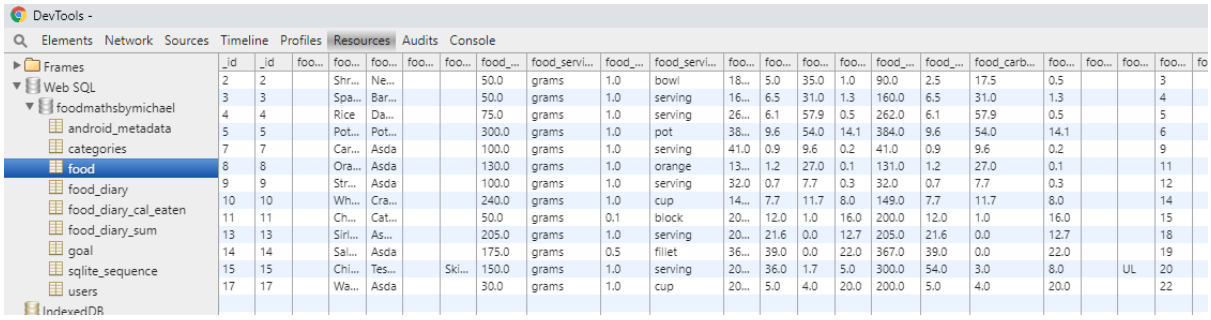
Fig. 12 (How to find the Android Version Distribution statistics in Android Studio, 2020)

When developing the GUI for my application a feature that really stood out was Android Studio's drag and drop capability, this feature allows developers to create user interfaces without writing out HTML code. Which meant I was able to spend much less time writing the

code for the GUI than I had previously planned on. This allowed me to design a much more comprehensive and intuitive user-friendly interface than I previously thought possible with my current ability.

4.2 Stetho

During the development of my artefact I was required to design and implement a database, this was made much simpler by the addition of Stetho, Stetho is an open source debugging tool for android development, which allows developers to use Chrome dev tools. I mainly used this tool for testing and debugging my database to ensure my artefact was adding, deleting and editing database entries when it was asked to. To say this piece of software was useful would be an understatement, it was the most critical factor in maintaining my database during development.



_id	_id	foo...	foo...	foo...	foo...	foo...	food_...	food_servi...	food_...	food_servi...	foo...	foo...	foo...	foo...	food_...	food_...	food_carb...	foo...	foo...	foo...	foo...	fo
2	2		Shr...	Ne...			50.0	grams	1.0	bowl	18...	5.0	35.0	1.0	90.0	2.5	17.5	0.5			3	
3	3		Spa...	Bar...			50.0	grams	1.0	serving	16...	6.5	31.0	1.3	160.0	6.5	31.0	1.3			4	
4	4		Rice	Da...			75.0	grams	1.0	serving	26...	6.1	57.9	0.5	262.0	6.1	57.9	0.5			5	
5	5		Pot...	Pot...			300.0	grams	1.0	pot	38...	9.6	54.0	14.1	384.0	9.6	54.0	14.1			6	
7	7		Car...	Asda			100.0	grams	1.0	serving	41.0	0.9	9.6	0.2	41.0	0.9	9.6	0.2			9	
8	8		Ora...	Asda			130.0	grams	1.0	orange	13...	1.2	27.0	0.1	131.0	1.2	27.0	0.1			11	
9	9		Str...	Asda			100.0	grams	1.0	serving	32.0	0.7	7.7	0.3	32.0	0.7	7.7	0.3			12	
10	10		Wh...	Cra...			240.0	grams	1.0	cup	14...	7.7	11.7	8.0	149.0	7.7	11.7	8.0			14	
11	11		Ch...	Cat...			50.0	grams	0.1	block	20...	12.0	1.0	16.0	200.0	12.0	1.0	16.0			15	
13	13		Siri...	As...			205.0	grams	1.0	serving	20...	21.6	0.0	12.7	205.0	21.6	0.0	12.7			18	
14	14		Sal...	Asda			175.0	grams	0.5	fillet	36...	39.0	0.0	22.0	367.0	39.0	0.0	22.0			19	
15	15		Chi...	Tes...		Ski...	150.0	grams	1.0	serving	20...	36.0	1.7	5.0	300.0	54.0	3.0	8.0		UL	20	
17	17		Wa...	Asda			30.0	grams	1.0	cup	20...	5.0	4.0	20.0	200.0	5.0	4.0	20.0			22	

Fig. 13. Food Database as seen by going to **chrome://inspect/** *while using Stetho*

5. Application Design

My intension when designing my application was to maintain a simplistic, almost minimalistic aesthetic, while ensuring the user would be able to understand what was on the screen and how they could interact with each element of the interface. The colours I chose were Blue and White, this is inline with similar applications on the market, there is some merit behind this, as Blue is widely known to symbolise Calm and Intelligence. In comparison when looking at fast food logos and imagery you will often find yellows and reds to be prevalent, blue is a colour that doesn't often take centre stage. For this reason, I have opted for the colour blue as my main theme as my application is not for the purpose of encouraging or discouraging the consumption of food, it is merely a tool to be used by its users to aid them with their goals.

5.1 Sign up Pages

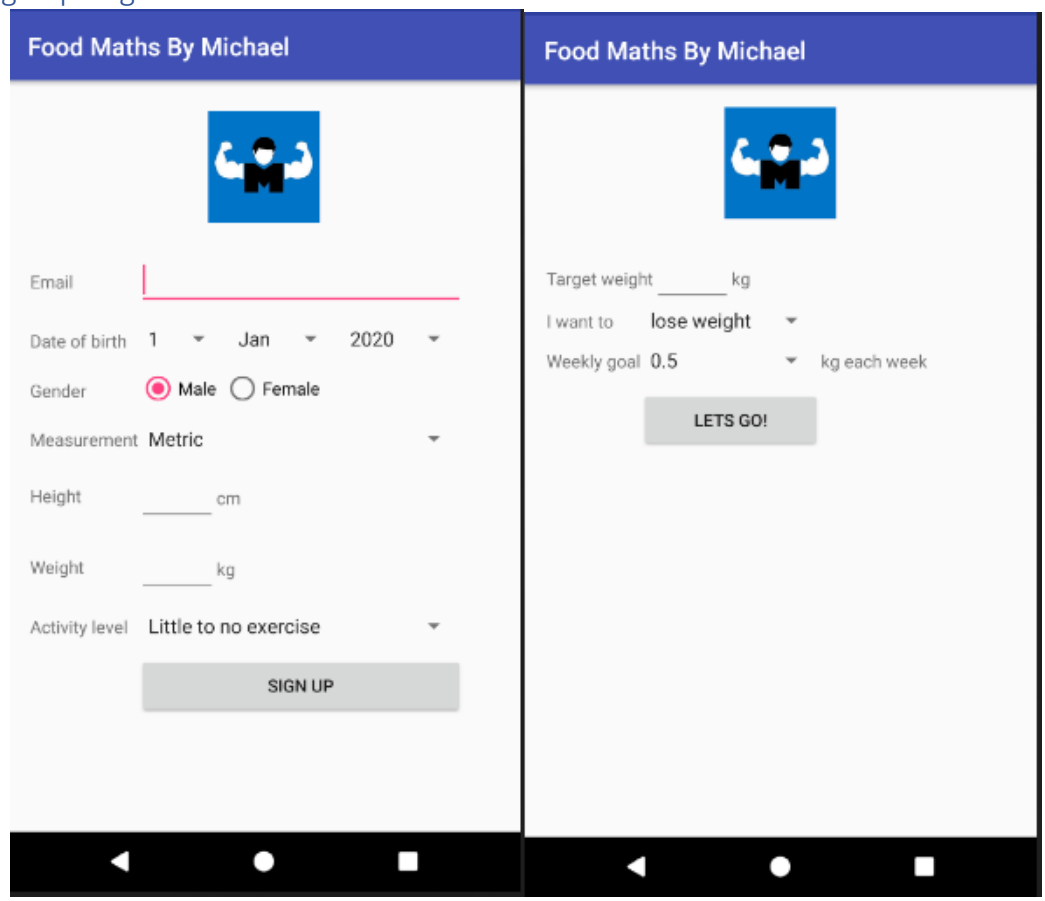


Fig. 14.

Upon starting the FoodMaths application for the first time the user is greeted by the first screen on the left, they are prompted to input their personal details. If the user presses the Sign-Up button without filling in the fields, they will receive an error message instructing them to complete the missing field(s). The information that is gathered on this first page is stored in the user table in the database; this data is also used to calculate the users daily caloric intake.

The second screen on the right is seen by the user after pressing the Sign-up button, this screen allows users to set they goal weight, and decide how much they intend to either lose or

gain each week. This information is also stored in the user table in the database and will be used when calculating the user's daily calorie intake goal.

5.2 Home Page

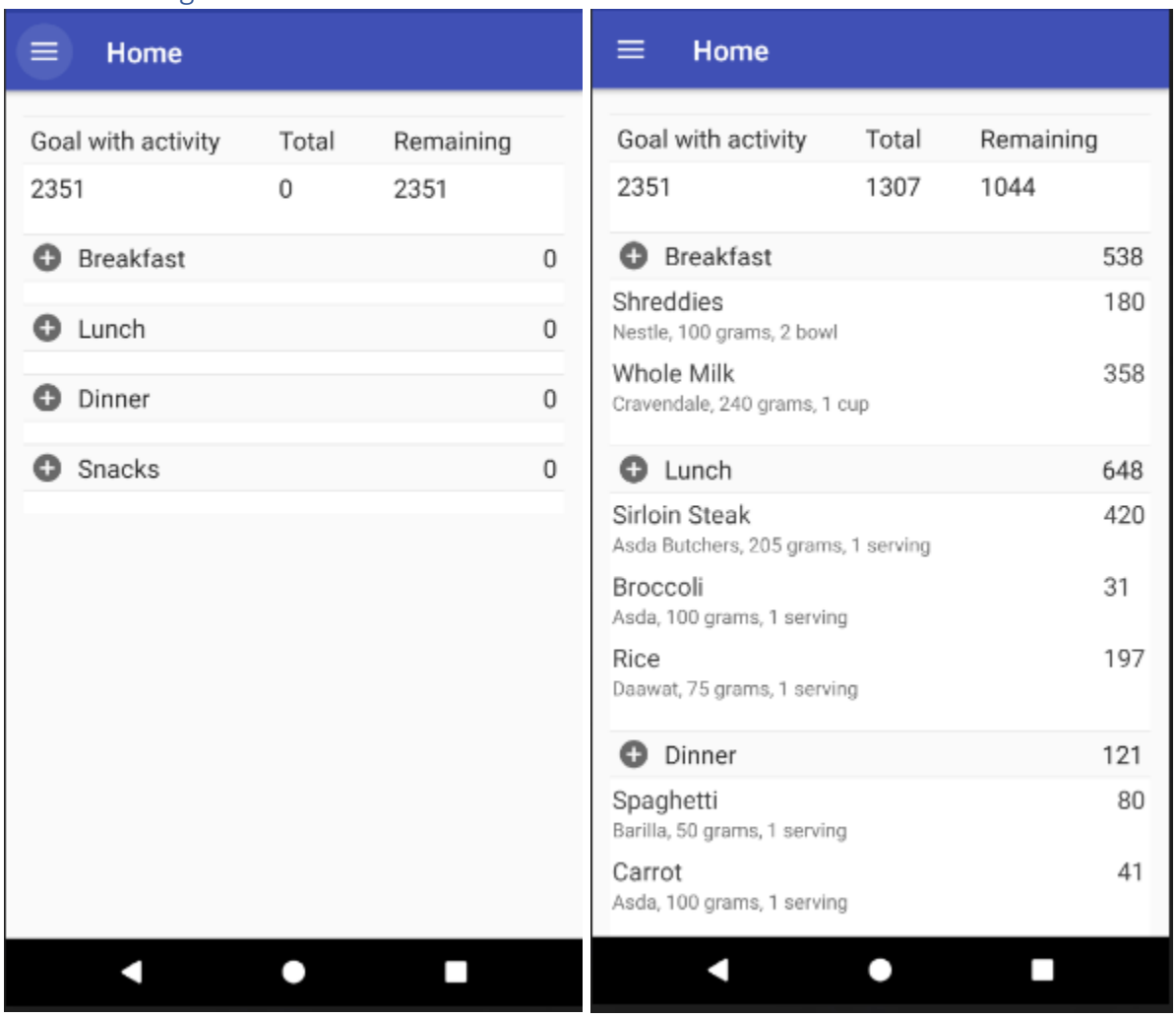


Fig. 15.

Once the user has completed the sign up, they will be redirected to their home page, this is the food dairy page. The two screens above show an empty food diary and that same food diary once it has been added to. At the top of the page the user's daily calorie goal is listed as well total consumed calories and remaining calories. By pressing on the '+' button under each of the meal names, the user can choose which foods to add to their diary and how much of each food they would like. Once food has been added the total number of calories consumed and remaining is recalculated, to provide the user with an up to date view of their daily calorie intake.

5.3 Profile Page

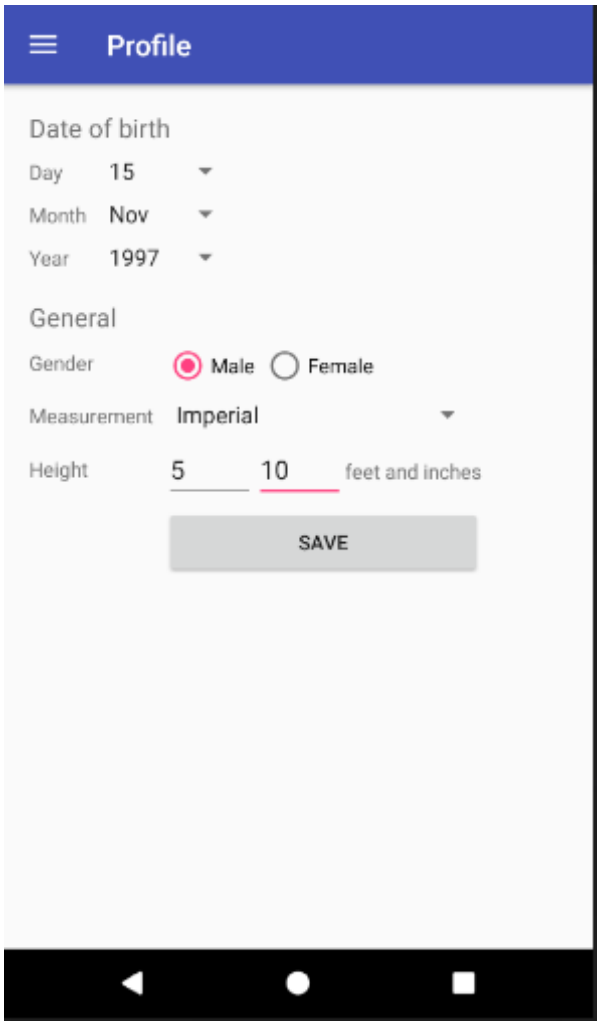


Fig. 16.

The profile page displays the user's personal information, this information can be altered by the user if they so wish. Once changed the user can save the changes, if the user then revisits the profile page their new details will be displayed. The user is also able to change between metric and imperial units for height.

5.4 Goal Page

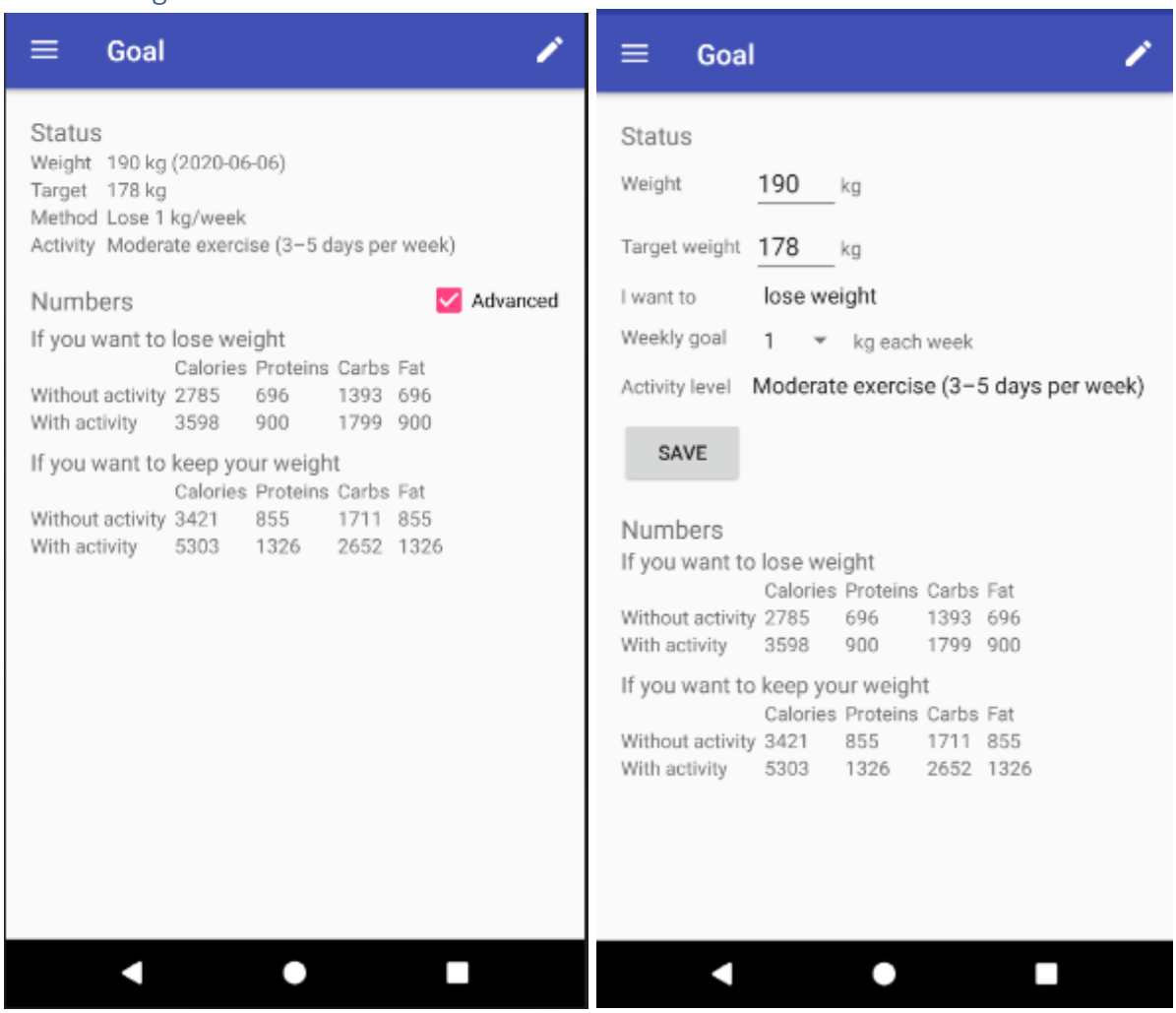


Fig. 17.

The goal page is used to display the users goal, the users current weight is listed along with the date that weight was recorded. Along with the users target weight and in this users case their weightloss plan and level fo activity. The second table below gives the user information based on their level of activity and informs the user of what might be a suitable calorie intake for them to achieve their desired goal.

An extra feature I have added is the tickable box which allows users to see an advanced breakdown of their total calorie intake goal into the three different macro nutrtients. This is not a necessary piece of information for most casual users, however many proffesional athletes will split their meals into these categories and this would help them keep a better track.

The second screenshot shows the edit goal screen which allows users to make any changes to their goal and current weight or activity level. Any changes that are made will cause the application to recalculate the users goal calorie intake to achieve their weight goal.

5.5 Food Page

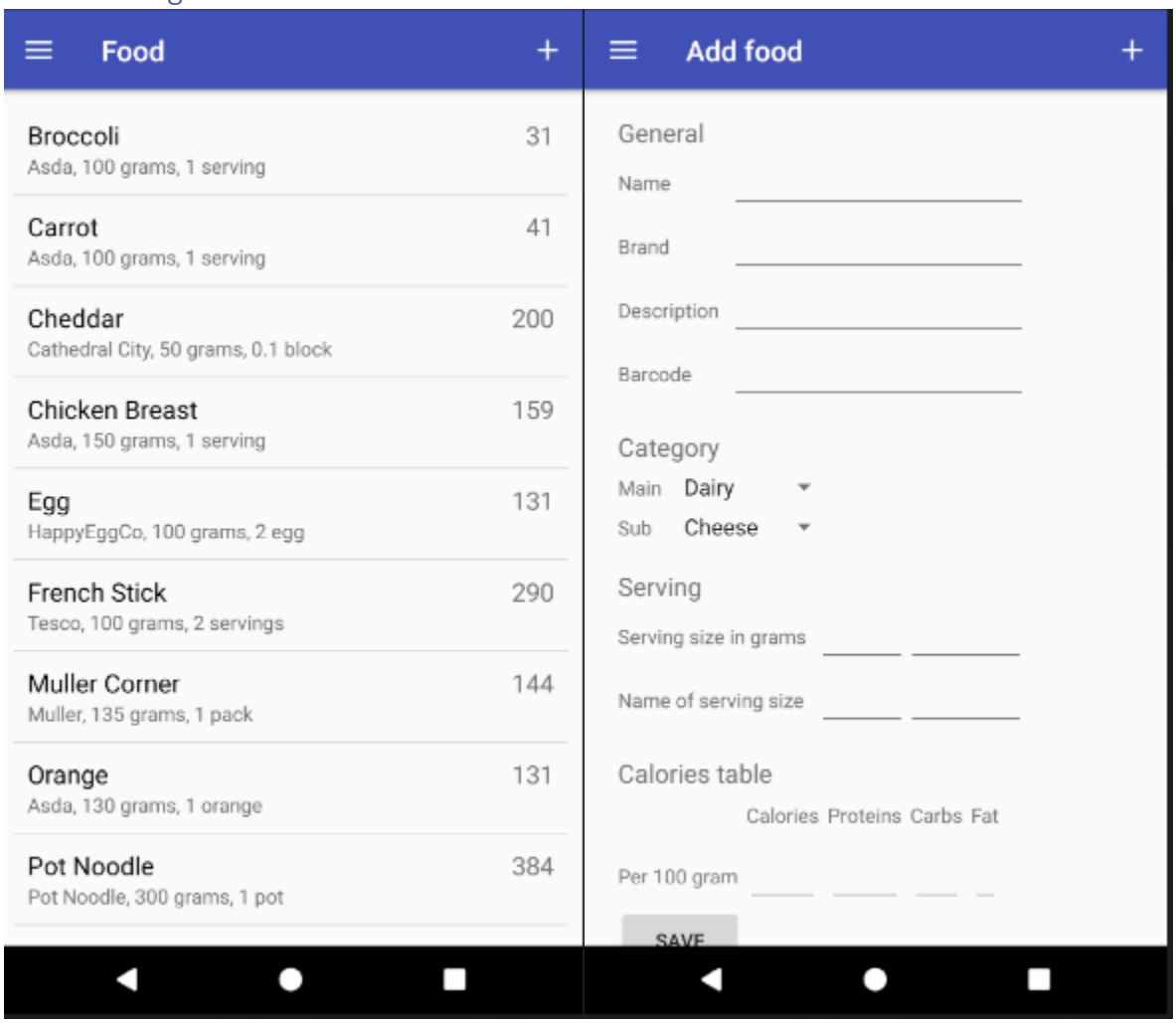


Fig. 18.

The final page is the food page which lists all the food items in the food table in the application database, listing foods in alphabetical order. The details shown are the foods name, brand, serving size and calories per serving on the far right.

If the user would like to add a new food to the database table, they must click the '+' at the top right of the screen and be redirected to the add food screen as shown on the right. In this screen the user can add any food item they wish, with all its own specific serving sizes and calories. Once this has been done the food item can be found from the home page and added to the food diary.

5.6 Navigation Bar

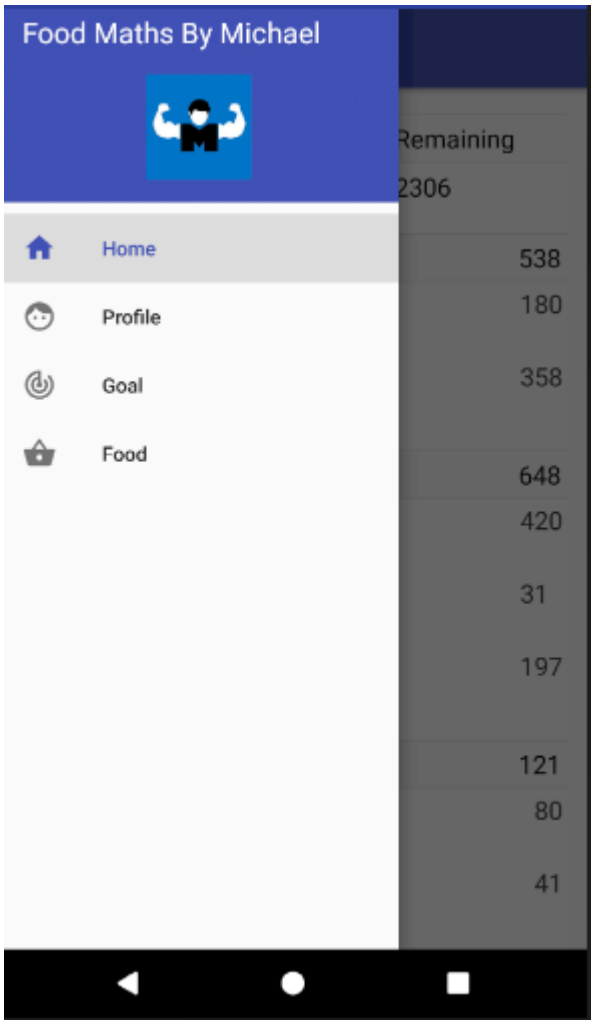


Fig. 19.

To navigate through my FoodMaths application I have designed a simple navigation bar, users simply click the 3 horizontal lines in the top left of the screen and the nav bar opens. From the nav bar users can move through the different screens of my application with ease.

6. Implementation

In this section of my report I will cover how my artefact was implemented and each of the different systems that came together to create the complete artefact.

6.1 Database Implementation

Prior to implementing my database, it was necessary for me to create an ER diagram to show the relationship between each of the tables my database will contain. This ER diagram had multiple stages to its creation, the result can be seen in the appendix 11.1

Once the ER diagram was finished, I had to begin by creating the DBAdapter (database adapter) class, this class is used to manage the entire database and all the functionality associated with the database i.e. inserting, selecting and deleting entries. This class also contains the private class DatabaseHelper which inherits all the functionality and methods of the SQLiteOpenHelper class.

```
public class DBAdapter {

    /* 01 Variables ---------------------------------------- */
    private static final String databaseName = "foodmathsbymichael";
    private static final int databaseVersion = 9;

    /* 02 Database variables ------------------------------- */
    private final Context context;
    private DatabaseHelper DBHelper;
    private SQLiteDatabase db;


    /* 03 Class DbAdapter ---------------------------------- */
    public DBAdapter(Context ctx){
        this.context = ctx;
        DBHelper = new DatabaseHelper(context);
    }

    /* 04 DatabaseHelper ----------------------------------- */
    private static class DatabaseHelper extends SQLiteOpenHelper {
        DatabaseHelper(Context context) { super(context, databaseName, null, databaseVersion); }
```

Fig. 20. Shows a snippet of the DBAdapter class

The DatabaseHelper class is used to initialise all the tables into the database, this includes the user table, goal table, food table, categories table, food diary table as well as all their individual column names. These tables and column names are all declared in the DatabaseHelper class.

```
try{
    // Create tables
    db.execSQL("CREATE TABLE IF NOT EXISTS users (" +
            " _id INTEGER PRIMARY KEY AUTOINCREMENT, " +
            " user_id INTEGER, " +
            " user_email VARCHAR," +
            " user_dob DATE, " +
            " user_gender INT, " +
            " user_height INT, " +
            " user_note VARCHAR);");

}
```

Fig. 21. Shows the user table being created in the DatabaseHelper class onCreate method

As shown in Figure 21 the onCreate method is called to create all of the tables, in this figure user table is instantiated with its column names, if the tables are ever altered in this class the databaseVersion INT in Figure 20 must be increment by 1, this will call the OnUpgrade method which drops all of the tables in the database and allows the database to be rebuilt with the new tables and column names.

```
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {


    // ! All tables that are going to be dropped need to be listed here

    db.execSQL("DROP TABLE IF EXISTS goal");
    db.execSQL("DROP TABLE IF EXISTS users");
    db.execSQL("DROP TABLE IF EXISTS food_diary_cal_eaten");
    db.execSQL("DROP TABLE IF EXISTS food_diary_sum");
    db.execSQL("DROP TABLE IF EXISTS food_diary");
    db.execSQL("DROP TABLE IF EXISTS categories");
    db.execSQL("DROP TABLE IF EXISTS food");
    onCreate(db);

    String TAG = "Tag";
    Log.w(TAG, "Upgrading database from version " + oldVersion + " to "
            + newVersion + ", which will destroy all old data");

} // end public void onUpgrade
```

Fig. 22. Shows OnUpgrade method

Once the tables had been created the next task was to fill them with some default data which I could use for testing purposes, this led to the creation of the DBSetupInsert (Database Setup Insert) class. This class fills the food table and categories table with data which can then be used on the application to add to a food diary, these tables can also be added to by users who wish to create their own food.

```java
public void setupInsertToFood(String values) {

    try {
        DBAdapter db = new DBAdapter(context);
        db.open();
        db.insert("food",
                "_id, food_name, food_manufactor_name, food_serving_size_gram, " +
                        "food_serving_size_gram_mesurment, food_serving_size_pcs, " +
                        "food_serving_size_pcs_mesurment, food_energy, food_proteins, " +
                        "food_carbohydrates, food_fat, food_energy_calculated, " +
                        "food_proteins_calculated, food_carbohydrates_calculated, " +
                        "food_fat_calculated, food_user_id, food_barcode, food_category_id, " +
                        "food_thumb, food_image_a, food_image_b, food_image_c, food_notes",
                values);
        db.close();
    } catch (SQLiteException e) {
        // Toast.makeText(context, "Error; Could not insert food.", Toast.LENGTH_SHORT).show();
    }

}
```

Fig. 22. Shows setupInsertToFood method

```java
public void insertAllFood() {
    setupInsertToFood("NULL, 'French Stick', 'Tesco', '100', 'grams', '2', 'servings', '290'," +
            " '9.6', '60', '3', '290', '9.6', '60', '3', NULL, NULL, '2', NULL, NULL, NULL, " +
            "NULL, NULL");
    setupInsertToFood("NULL, 'Shreddies', 'Nestle', '50', 'grams', '1', 'bowl', '180', '5', " +
            "'35', '1', '90', '2.5', '17.5', '0.5', NULL, NULL, '3', NULL, NULL, NULL, NULL, " +
            "NULL");
    setupInsertToFood("NULL, 'Spaghetti', 'Barilla', '50', 'grams', '1', 'serving', '160', " +
            "'6.5', '31', '1.3', '160', '6.5', '31', '1.3', NULL, NULL, '4', NULL, NULL, " +
            "NULL, NULL, NULL");
```

Fig. 23. Shows insertAllFood method

In Figure 22 and Figure 23, you can see how the food data has been inserted into the database, setUpInsertToFood setups the database so I can then add each individual food item and its fields. This is done as seen in insertAllFood where I must manually input the food data fields. As can be seen a few of the fields are NULL this is due to me adding a few extra fields which are not currently in use throughout my artefact now such as the food images. I had created these at the start just to be sure that if I wanted to add them in time, I would be able to and save myself the work of going back and changing the all the tables. I have kept these fields there as I believe in the future I may come back and add improvements to my artefact.

6.2 Application Implementation

Once the database had been completely implemented into the system, I then began with the application implementation. The reason I left this until after the database was implemented, is due to the fact that the database is needed throughout the application implementation phase, for testing how the application interacts with the data contained within the database, whether it can perform all of its actions correctly.

6.2.1 Sign Up and Sign Up Goal

The Sign-up activity was the first activity I chose to focus on as it is a chance for the application to gather user data which can later be used and displayed throughout the other

activities. This activity in large serves the main purpose of gathering user details, however on top of this there is also a large portion of this activity dedicated to giving the user error messages if they have missed out certain fields.

```
/*- Sign up Submit ---------------------------------------------- */
public void signUpSubmit() {
    // Error
    ImageView imageViewError = (ImageView)findViewById(R.id.imageViewError);
    TextView textViewErrorMessage = (TextView)findViewById(R.id.textViewErrorMessage);
    String errorMessage = "";

    // Email
    TextView textViewEmail = (TextView)findViewById(R.id.textViewEmail);
    EditText editTextEmail = (EditText)findViewById(R.id.editTextEmail);
    String stringEmail = editTextEmail.getText().toString();
    if(stringEmail.isEmpty() || stringEmail.startsWith(" ")){
        errorMessage = "Please fill inn an e-mail address.";
    }
```

Fig. 24. Shows the code for SignUpSubmit in the SignUp class

As can be seen in Figure 24 the method singUpSubmit is triggered once the user presses the sign-up button on the sign-up screen. Once this has been clicked the method checks if any of the user details fields have been left blank, if any fields are not complete the application will return an error message to the user prompting them to input the correct details and make the necessary changes.

As can be seen is Figure 25 the user has pressed the submit button but not been able to proceed as the height field has not bee filled in, the application will keep the user on this page and display a clear error message with a description of why the user has not been able to progress further. Once the user fills in the field, they can press the sign-up button and be taken to the next page the sign-up goal page.

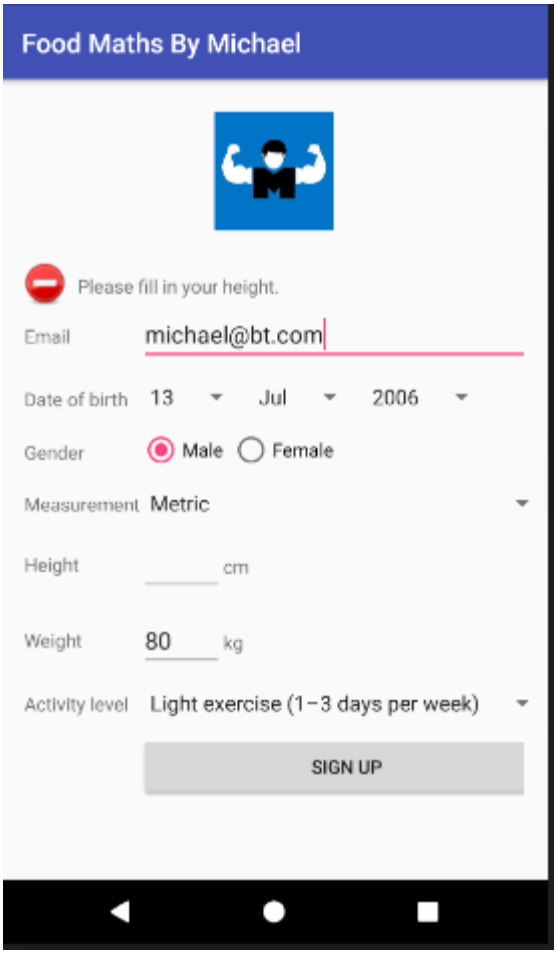


Fig. 25. Shows the sign-up screen with error message.

The sign-up goal page works similarly to the sign-up page in that it is mainly used to gather user details this time the users target weight and weekly weight loss goal are the details taken. However, the sign-up goal activity is much more complex as it handles all the mathematical calculations behind the users' weight loss plan. This activity uses the users input data (age, height, gender, current weight and weekly weight loss goal) to calculate the users daily calorie intake goal, this was a long process to setup as the there are a lot of factors which can alter this figure and they all needed to be accounted for in the calculations.

```
/* 1: BRM */
// Start calculation
double goalEnergyBMR = 0;
if (stringUserGender.startsWith("m")) {
    // Male
    // BMR = 66.5 + (13.75 x kg body weight) + (5.003 x height in cm) - (6.755 x age)
    goalEnergyBMR = 66.5 + (13.75 * doubleUserCurrentWeight) + (5.003 *
            doubleUserHeight) - (6.755 * intUserAge);
    //bmr = Math.round(bmr);
    //Toast.makeText(this, "BMR formula: 66.5+(13.75*" + doubleUserCurrentWeight + ")
    // +(5.003*" + doubleUserHeight + ")-(6.755*" + intUserAge + " = " +
    // goalEnergyBMR, Toast.LENGTH_LONG).show();

} // if(stringUserGender.startsWith("m")){
else {
    // Female
    // BMR = 55.1 + (9.563 x kg body weight) + (1.850 x height in cm) - (4.676 x age)
    goalEnergyBMR = 655 + (9.563 * doubleUserCurrentWeight) + (1.850 *
            doubleUserHeight) - (4.676 * intUserAge);
    //bmr = Math.round(bmr);
}
```

Fig. 26. Shows the code to work out the user BMR(base metabolic rate)

As shown in Figure 26 the application works out the users BMR, this figure is applied throughout the sign-up goal page, alongside the user's level of activity and weekly weight loss target to come up with a figure for each specific user's daily calorie goal. Once this has been achieved this figure is then displayed in the food diary so that the user is able to keep track of how they well they are doing. If at any point the user wishes to change their details, they may do so on the goal page, this will prompt the application to redo these calculations and come up with a new BMR and daily calorie intake goal for the user.

6.2.2 Home

The home fragment deals with the food diary page this is the most sophisticated of the fragments in my artefact as it must interact with the food diary table, user table and the food table. The home fragment displays to users their current daily food diary and the breakdown of the foods in the diary, allowing users to insert, delete and edit their diary entries. Fragments are a combination of an XML layout file and a java class similar to activities however they can be designed to be reusable with logic embedded in them. In my case I have decided to use a fragment as it allows me to keep the same layout without creating multiple layout files. This is useful as in the case of my application users will often be adding, editing and deleting diary entries and it would make very little sense to design a specific diary layout file for each of the possible conditions. Thus, this saves me time and creates a more streamlined development process.

```
// Update fdce
if (stringMealNumber.equals("0")) {
    TextView textViewEnergyX = (TextView) getActivity().findViewById(R.id
            .textViewEnergyBreakfast);
    textViewEnergyX.setText("" + intFdceEatenEnergy);
} else if (stringMealNumber.equals("1")) {
    TextView textViewEnergyX = (TextView) getActivity().findViewById(R.id
            .textViewEnergyLunch);
    textViewEnergyX.setText("" + intFdceEatenEnergy);
} else if (stringMealNumber.equals("4")) {
    TextView textViewEnergyX = (TextView) getActivity().findViewById(R.id
            .textViewEnergyDinner);
    textViewEnergyX.setText("" + intFdceEatenEnergy);
} else if (stringMealNumber.equals("5")) {
    TextView textViewEnergyX = (TextView) getActivity().findViewById(R.id
            .textViewEnergySnacks);
    textViewEnergyX.setText("" + intFdceEatenEnergy);
}
```

Fig. 27. Shows the home fragment creating the food diary menu table with each meal name

The home fragment works by checking the food diary table which contains the food diary ids, for the correct food. The home fragment then displays the changes for the user to see.

```
// Update fd energy calculated
double doubleFdEnergyCalculated = Math.round((doubleFdServingSizeGram *
        doubleGetFromSQLFoodEnergy) / 100);
String stringFdEnergyCalcualted = "" + doubleFdEnergyCalculated;
String stringFdEnergyCalcualtedSQL = db.quoteSmart(stringFdEnergyCalcualted);
db.update("food_diary", "_id", longFdID, "fd_energy_calculated",
        stringFdEnergyCalcualtedSQL);
```

Fig. 29 shows the food diary energy(calories) being updated after the users makes a change

The home fragment also handles all the additional mathematical calculations which are needed to provide the user with a breakdown of their own daily calorie and food intake. As shown in Figure 29 the total energy consumed is being updated, this will then be displayed for the user to see on their food diary screen.

6.2.3 Profile

The profile fragment is used to display users' personal details and data, this is achieved by sending a query to the user table and retrieving the specific user data. This profile fragment is also used to allows users to make any alterations to their profile that they desire, if the user needs to update their current weight or age, they would do this by visiting the profile fragment. When the user presses the save button, the new details are saved and the old data is deleted, this is done by calling the onUpdate method which updates the fields inside the user table just as it was done during the initial user signup when the users data was first entered into the table.

6.2.4 Goal

The goal fragment is used to display a users' weight goal and a full breakdown of the user's calorie intake along with macronutrients. The fragment also lets users edit and save new goals, this is done in the same way as the profile fragment handles its update, by calling the onUpdate method and editing the fields in the goal table. Once this has been saved by the user, they can reload the fragment and will be able to see their updated goal.

6.2.5 Food

The food fragments purpose is to display all the foods from the food database and allow users to edit any of the already created foods and create their own new food. Implementing this fragment was a long process as there are many fields in the food table which need to be handled by the fragment. I have attempted to keep the display of information simple enough for everyday users to understand.

```
/* Get data from fragment */
Bundle bundle = this.getArguments();
if(bundle != null){
    currentId = bundle.getString("currentFoodId");

    // Need to run to get edit and delete buttons:
}
if(currentId.equals("")) {
    // Populate the list of categories
    populateListFood();
}
else{
    preListItemClickedReadyCursor(); // We are com
}
```

Fig. 30. Shows the food fragment retrieving food data to display as a list

7. Testing

Software testing is done to ensure code meets the requirements that were set out at the beginning of the development plan, in this case I will be testing against my project proposal requirements which are in Figure 9 and 10. I will be using Unit testing to see how the system acts when comparing the actual system outcome to an expected correct outcome.

7.1 Unit Testing

Test No	Name	Type	Expected Outcome	Actual Outcome	Pass/ Fail
1	User sign up	Functional	Users can complete the sign-up process	Users can complete the sign-up process	PASS
2	User sign up goal	Functional	Users can choose a goal	Users can choose a goal	PASS
3	Edit user profile	Functional	Users can edit their details	Users can edit their details	PASS
4	Edit user goal	Functional	Users can edit their goal	Users can edit their goal	PASS
5	Create new food	Functional	Users can add new foods to the database	Users can add new foods to the database	PASS
6	Edit food	Functional	Users can edit food in the database	Users can edit food in the database	PASS
7	Delete food	Functional	Users can delete foods from the database	Users can delete foods from the database	PASS
8	Add food to diary	Functional	Users can add foods from the database into their diary	Users can add foods from the database into their diary	PASS
9	Edit food in diary	Functional	Users can edit the food in their diary	Users can edit the food in their diary	PASS
10	Delete food in diary	Functional	Users can delete the food in their diary	Users can delete the food in their diary	PASS
11	Display calories eaten	Functional	Users can see a display of their total calories eaten	Users can see a display of their total calories eaten	PASS
12	Display calories remaining	Functional	Users can see a display of their total calories remaining	Users can see a display of their total calories remaining	PASS
13	List foods by categories	Functional	Users can list foods by categories	Users are unable to list foods by categories; however, they	MARGINAL FAIL

				can view the categories from the home screen when adding foods to their diary	
14	Create new food categories	Functional	Users can create new food categories	Users are unable to create new categories	FAIL
15	Null fields on sign up or sign up goal	Null	Users are unable to proceed if they leave fields blank	Users are unable to proceed if they leave fields blank.	PASS
16	User error messages	Non-functional	Users are provided with a message if they encounter a bug	Users are provided with a message if they encounter a bug	PASS
17	Operate across a wide range of devices	Non-functional	The device runs on Oreo 8.0 for android	The device runs on Oreo 8.0 for android	PASS
18	Responsiveness	Non-functional	Artefact should be quick to respond to inputs	Artefact is quick to respond to inputs, almost not delay	PASS
19	State	Non-functional	Users should be informed by the artefacts current state	Using visual cues such as load bars the artefact does inform users of its current state	PASS
20	Database size	Non-functional	The database should be able to accommodate many user additions	The database can handle many new entries in all its tables	PASS
21	Navigation	Functional	The user should be able to navigate effectively through the artefact using simple buttons	The user can navigate effectively through the artefact using simple buttons	PASS

Fig. 31. Shows Unit testing results

As can be seen in Figure 31 the results of the Unit testing are overall very positive most of the functionality has been completed as expected in the two cases that were not successful, I will elaborate why.

Test 13 was a marginal failure as users are not able to list foods by their food categories in their own separate fragment page, this would have been the categories fragment, this was how I had previously intended on this feature working. After some time spent working on

this, I was unable to get it working to a good standard, instead I opted for implementing this feature directly into the home fragment and food diary. Now when users click to add a new food, they can browse all the foods in the database by categories, this is different to how the foods appear in the food fragment as there they are displayed in alphabetical order by their name.

Test 14 failed as once I had decided against adding the category fragment, I no longer wished to add user created categories and instead opted to create a wide range of default categories manually and have these saved into the database and accessible to all users. In some ways this negates the need for the user created categories as the categories are simply a filtering tool used to find specific food, once large variety have been made there should be no need to make any additional categories.

7.2 Usability Testing

For my project I was unable to undergo any usability testing as I would have required ethics approval which was not submitted. Nevertheless, usability testing would have been a great way of acquiring useful feedback which I could have then implemented back into developed and made any of the necessary changes to improve my artefact for the better. If I was to take my artefact development further, I would consider the prospect of usability testing as the benefits it can bring to app development are high. "The mathematical benefits of adding test users should be cited. More test users means greater confidence that the problems that need to be fixed will be found" (Faulkner, 2003)

8. Evaluation

In this chapter I will be covering what I have achieved during the development of my artefact and evaluating the entire project to see whether each objective was achieved and provide a detailed look into this, discussing the challenges faced and how they were overcome.

8.1 Project Aim and Objectives

My aim as stated in section 1.2 of this report has to the best of my ability been complete, I have been able to produce a fully functioning fitness and diet tracking application, that is able to provide some of the main functionality of current applications which are available on the market. I have achieved this by building the base functionality and adding intuitive user-friendly design in the form of menus and navigation options.

From my project objectives stated in section 1.3 I have shown that I am able to apply and develop the concepts I have learnt throughout the software engineering modules and produce an artefact that not only performs well, it also aids in addressing a global need/want. I have also applied material learnt from previous years modules 5COM1052 Database Concepts, to design and implement a working database with the ability to handle SQL queries and I have integrated this into my application.

As I stated in my objectives, I was aware of the possible issues that may have arisen from time constraints, this however did not prevent me from completing a working artefact. Even though my artefact is far from market ready it is functional, and users can till benefit from its capabilities. With further development my artefact could be a competitor with the top applications available.

8.2 Challenges faced

The main challenge faced throughout the development of my artefact was the challenge of managing, implementing and testing the entire project by myself. This was the first time I have undertaken a project of this size and without the direct assistance of others in the development process it proved to be a very time consuming and at times infuriating process. This way I overcame this challenge was by keeping a diary containing all the tasks that I needed to complete, therefore staying time aware and also by doing lots of additional learning which facilitated my skill development helping me to complete tasks independently for which I would have previously required assistance.

Then I faced the challenge of implementing a database containing multiple tables and designing how each of the tables would link with each other using ERDiagrams to visual this. I was starting this task with a brief knowledge base acquired from previous studies, however I soon realised I would need to undertake some more research and acquire a wider breadth of knowledge to design the database the way I had intended to. (Tamada, 2020) (SQLiteOpenHelper | Android Developers, 2020)

The final challenge faced was the design of the fragment classes which could display varying information without having to redesign a completely new layout page. Before starting the development, I had not planned on adding fragment classes, however after further research I had concluded this would provide my application with an additional level of complexity which would allow me to improve my software engineering skills and expand my knowledge base. (Fragments | Android Developers, 2020)

8.3 Future Work

Although after my artefact was complete and I am very satisfied with the standard, there was some functionality which was omitted from the final product. If I were given extra time to continue development, I would have added the categories fragment and menu button, which as stated in section 7.1 would have given users the capability to sort food by category i.e. dairy, meat or vegetables. This feature would have aided in simplifying and keeping my application minimalistic which was an additional goal I was striving for.

Given further development time and a greater skill set the addition of social features would have been a possibility, allowing users to share their food diary with friends and interact with others posts. Implementation of accessibility features meaning users could edit font sizes and change colour palettes would aid in a greater usability and expand the possible user base of my application.

A deeper breakdown into the user's statistics using graphical representation would be a feature I intend on developing if the opportunity became available to do so. This would appeal to more professional users, those who wish to monitor and compare their daily stats, with the utmost accuracy and precision.

8.4 Conclusion

To conclude the development of this project has been challenging, at times I ran into bugs which caused a complete stand still in development, particularly during the implementation of the database. However, after looking back at my statement of success from my project proposal I can say that I have personally succeeded in following the criteria I had set out. From the knowledge I have gained I have managed to develop my software engineering skills which will facilitate me in my future life endeavours.

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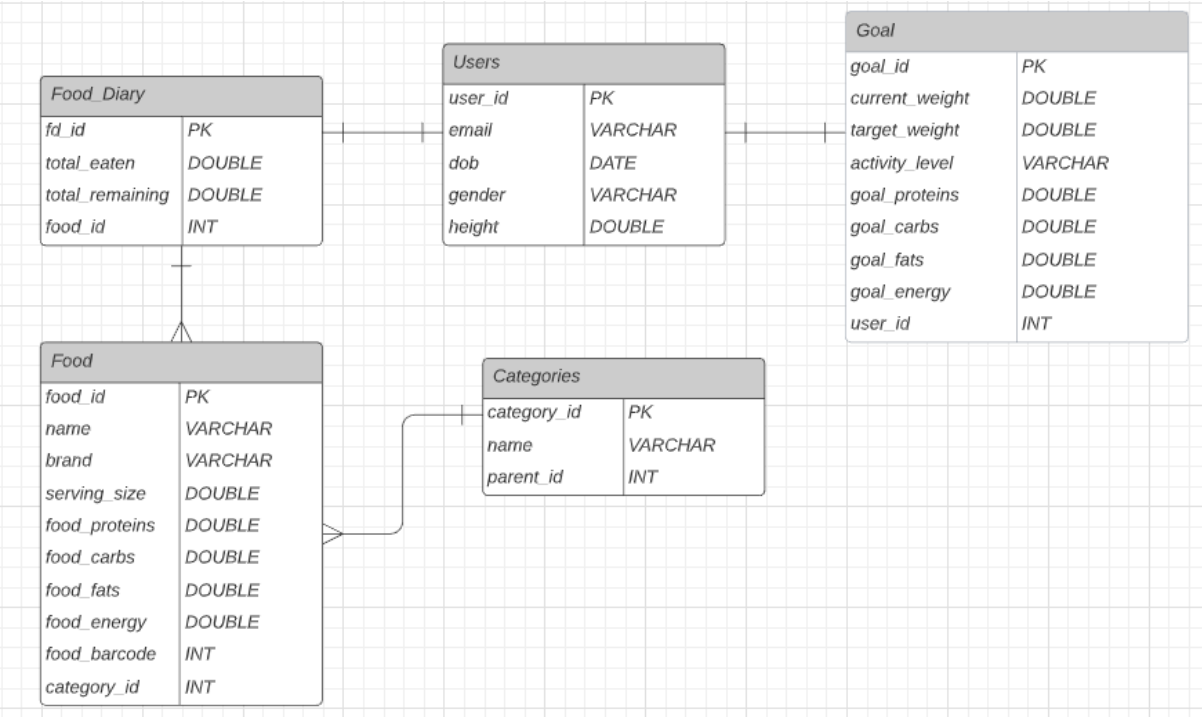
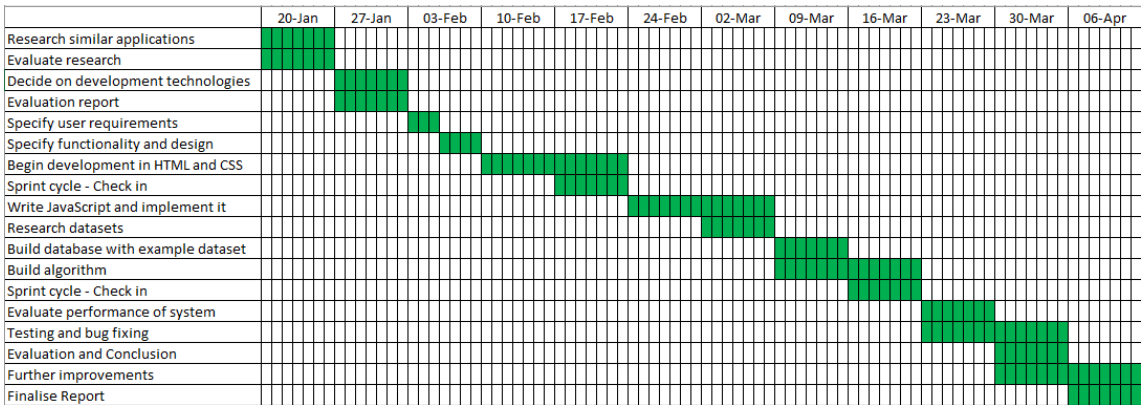
10. Appendix

10.1 ER Diagram



10.2 Gant Chart